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Studierichting: Informatica

Oefeningenreeks 4A nr. 61.4

$$\int \frac{1}{x^3 \sqrt{x-1}} dx \stackrel{t=\sqrt{x-1}}{=} \int \frac{1}{(t^2+1)^3} \cdot 2t dt = 2 \int \frac{1}{(t^2+1)^3} dt = I$$

$$\Rightarrow M_1 = \text{Bgtan } t + c$$

$$\Rightarrow M_2 = \frac{1}{2} \text{Bgtan } t + \frac{t}{2(t^2+1)}$$

$$\Rightarrow M_3 = \frac{3}{4} \left(\frac{1}{2} \text{Bgtan } t + \frac{t}{2(t^2+1)} \right) + \frac{1}{4(t^2+1)^2} + c$$

$$I = \frac{3}{4} \text{Bgtan } t + \frac{3t}{4(t^2+1)} + \frac{t}{2(t^2+1)^2} + c$$

$$\stackrel{t=\sqrt{x-1}}{=} \frac{3}{4} \text{Bgtan } \sqrt{x-1} + \frac{3\sqrt{x-1}}{4x} + \frac{\sqrt{x-1}}{2x^2} + c$$

References